

13_Final Report

Executive Summary

The Olist Brazilian E-Commerce Performance Analysis project was conducted to evaluate marketplace performance across revenue generation, customer behavior, product performance, operational efficiency, and customer satisfaction.

Using Python, PostgreSQL, and Power BI, transactional marketplace data was transformed into actionable business insights. The analysis identified key revenue drivers, customer purchasing patterns, operational strengths, and growth opportunities that can support data-driven decision-making.

The project revealed strong revenue performance, efficient order fulfillment, and positive customer sentiment. However, customer retention emerged as the primary business challenge, with most customers making only a single purchase.

Project Objective

The objective of this project was to analyze Olist's marketplace performance and answer critical business questions related to:

- Revenue growth
- Customer acquisition and retention
- Product category performance
- Operational efficiency
- Customer satisfaction

The analysis aimed to identify business risks, growth opportunities, and strategic recommendations for long-term marketplace success.

Dataset Overview

Dataset: Olist Brazilian E-Commerce Public Dataset

Core Business Entities:

- Customers
- Orders
- Order Items
- Payments
- Products
- Sellers
- Reviews
- Category Translation

Analysis Period:

- September 2016 to August 2018

Tools Used:

- Excel
- Python
- PostgreSQL
- Power BI

Methodology

The project followed a structured analytics workflow:

1. Business Understanding

Defined business objectives, stakeholder requirements, KPIs, hypotheses, and analytical framework.

2. Data Preparation

Validated data quality, checked missing values, verified relationships, and prepared data for analysis.

3. SQL Analysis

Developed KPIs, performed hypothesis testing, and conducted advanced SQL analysis using window functions and analytical techniques.

4. Power BI Development

Built a multi-page dashboard to visualize revenue, customer, product, operational, and satisfaction metrics.

5. Business Recommendation Development

Converted analytical findings into strategic recommendations and growth opportunities.

Key Performance Metrics

Metric	Value
Total Revenue	R\$16.01M
Total Customers	96K
Total Orders	99K
Average Order Value	R\$160.99
Average Review Score	4.09 / 5
Repeat Purchase Rate	3.12%
Delivery Success Rate	97%+

Major Findings

Revenue Performance

- Revenue exceeded R\$16 million during the reporting period.
- Revenue generation was concentrated within a small number of states.
- São Paulo generated the largest revenue contribution.
- Seasonal revenue patterns were observed throughout the analysis period.

Customer Performance

- Customer acquisition remained strong throughout the reporting period.
- Most customers completed only one purchase.
- Repeat purchasing behavior remained limited.

Product Performance

- Health & Beauty was the highest revenue-generating category.
- Bed Bath & Table generated the highest order volume.
- Revenue was concentrated among a small number of categories.

Operational Performance

- Delivery performance remained strong.
- More than 97% of orders were successfully delivered.
- Delivery delays affected a relatively small percentage of orders.

Customer Satisfaction

- Average review scores remained above 4.0.
- Five-star reviews represented the majority of customer feedback.
- Customer sentiment remained consistently positive.

Business Risks

1. Low customer retention.
2. Revenue concentration within a small number of regions.
3. Dependence on a limited number of product categories.
4. Delivery delays during peak demand periods.

Growth Opportunities

1. Improve repeat purchasing behavior through retention initiatives.
2. Expand successful product categories into additional markets.
3. Reduce revenue concentration by growing underperforming regions.
4. Leverage seasonal demand patterns for marketing and inventory planning.
5. Continue improving customer experience using review feedback.

Strategic Recommendations

Customer Retention

- Introduce loyalty and rewards programs.
- Develop personalized marketing campaigns.
- Create post-purchase engagement strategies.

Revenue Growth

- Expand investment in top-performing markets.
- Promote emerging product categories.
- Diversify revenue sources across regions.

Operational Excellence

- Improve delivery performance monitoring.
- Investigate recurring delay patterns.
- Optimize logistics planning during high-demand periods.

Customer Experience

- Analyze low-rated reviews regularly.
- Implement feedback-driven service improvements.
- Maintain customer satisfaction standards.

Final Conclusion

The analysis demonstrates that Olist operates a successful marketplace supported by strong revenue generation, efficient fulfillment operations, and positive customer satisfaction. While operational and customer experience metrics indicate a healthy business, customer retention remains the most significant opportunity for improvement.

Future growth can be accelerated by increasing repeat purchases, reducing revenue concentration risks, and expanding successful product categories into new markets. These initiatives can strengthen long-term customer value and support sustainable marketplace growth.